

AI Red Team Pen Testing Research Paper

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Artificial Intelligence is growing faster than ever, and it has now reached Cyber Red Teaming. Cyber Red Teaming is when an organization assesses how strong and vulnerable their organization is as a whole. This normally involves hiring a third-party to do it, but they assess the organization's staff, infrastructure, and operations against cyber threats. With Artificial Intelligence now being used in cyber red teaming there are multiple different ways that it can be used. Additionally, there are capabilities and limitations when implementing AI in cyber red teaming, as well as ethical and legal considerations. I will be discussing these topics in depth throughout this paper, along with the future possibilities with AI in cyber red teaming.

AI Driven Offensive Techniques

When it comes to the techniques and implementations of Artificial Intelligence within red teams, there are many different ways it can be used. One way that Artificial Intelligence is being implemented in society today is automated planning. State space planners, planning graph planners and hierarchical task network-based planners are the 3 main categories of automated planners (Yuen). State space planners are graphs that lay every state that the system could be in inside a node. These nodes are connected by arcs that represent an action to get to that state. Once the graph is constructed the AI then only needs to find the most suitable path (Yuen). This technique is only good with environments that are fully known and observable, as this would allow the planner to graph out the whole system. "Planning Graphs are data structures that capture both the set of possible and impossible states a system can be in by keeping track of mutexes: pairs of observations that cannot be true at the same time" (Yuen). This is similar to the State space planner, however with planning graphs there are two layers, an action layer and a propositional layer. The action layer is the possible actions that could be taken at a given step while the proposition layer represents the state of the system after any one of those actions are taken. These graphs extend until all the required propositions are true, this is the

shortest path. These graphs only grow polynomial and are often integrated in state space planners (Yuen). The last major category of planners is the hierarchical task network-based planners. Hierarchical Task Networks “are driven by the relationship between actions” (Yuen), rather than the state. To explain, hierarchical task networks take a highly complex task and break it down into smaller manageable subtasks called primitive tasks. These primitive tasks then have preconditions that need to be fulfilled to generate a valid plan. Hierarchical task networks are the most popular technique and are often used in robotics and NPC scripting in computer games (Yuen).

Another way that Red Teams can utilize Artificial Intelligence is by having them commit an attack on the organization themselves, an AI-powered cyber-attack. These can be useful in many ways to an organization as, “Red teaming using the AI-driven attacks could help the identification of vulnerabilities to novel techniques” (Al-Azzawi, 2025). One example of this Zouave et al, who employed Artificial Intelligence throughout different stages of his attack. He used AI to target “URLs, individuals’ personal data in search of relationships, passwords, captchas, and domains” (Al-Azzawi, 2025). This allowed the group to test the organizations automated detection system and social engineering vulnerabilities. They created deceptive URLs, hidden harmful links in conversations, solved captchas, created fake domains, and stole passwords. This allowed the organization to not only test their AI against a malicious AI, but also to point out their vulnerabilities.

Capabilities and Limitations

Artificial Intelligence has many advantages when it comes to red teams. One advantage that using artificial intelligence will give red teams is scalability and speed. “AI algorithms can perform vulnerability scans and threat simulations at a far greater pace than traditional methods” (CSA, 2024). These models are able to rapidly adapt to new environments and assess risks in real time, making them extremely useful in red teaming. Another advantage of AI in red teams is the adaptive learning aspect of AI. Artificial intelligence can quickly pick up trends and patterns from old data and apply those concepts

to the new environment. This can even allow them to predict new threats, as they constantly update their methodologies (CSA, 2024). In addition to these, artificial intelligence also has the ability to solve complex vulnerabilities that traditional tools typically cannot detect (CSA, 2024). These vulnerabilities can be user, system, and network behaviors. The last advantage that AI has when being implemented is the cost. Implementing AI can be costly itself, however, with the saved costs of time implementing it could reduce the overall cost.

With that being said, one limitation to artificial intelligence in red teams is the practical limitations. This includes automated tools generating false positives or false negatives, both of which require human intervention (EC-Council, 2024). This could be a huge limitation, as false negatives will leave vulnerabilities undetected and false positives will cause unnecessary interventions. Another limitation that comes with AI being implemented is their lack of contextual understanding. This makes it difficult for the AI to accurately understand complex scenarios. This again brings back the need for human intervention or collaboration. A limitation that implementing AI brought with it to cyber security is AI being in cybercrime. This created a continuing conflict between AI defense mechanisms and AI attacks. As new tools are being created for defense mechanisms, the same tool could be manipulated by criminals to evade the defense (Temidayo, 2025).

Ethical and Legal Considerations

When it comes to ethical and legal considerations, AI being used to penetrate a network can be ethically concerning. This is due to the fact that if the system got into the wrong hands, being able to autonomously penetrate networks could be a lot easier for criminals (CSA, 2024). This raises a need for strict regulations and oversights to ensure that the systems do not fall into the wrong hands. Another ethical consideration that comes with implementing artificial intelligence is accountability. This is due to the fact that human experts make informed and reasoned decisions, whereas AI is not always

transparent, because it relies on algorithms. “This lack of transparency complicates the process of validating security findings and addressing vulnerabilities” (Temidayo, 2025).

When It comes to the legal concerns, accountability is also an issue. This is due to the fact that if a system identifies a false positive, it could lead to operational consequences within the organization. The legal concern here is who is responsible, as it could be the developer, the operator, or the system itself (Jaiswal, 2024). Another Legal consideration is the potential for bias in the AI system’s algorithm. This can be in the form of “disproportionately targeting specific individuals or entities based on flawed data assumptions” (Jaiswal, 2024). This can lead to discrimination which could be a direct violation of anti-discrimination laws. In addition to these, another legal consideration is with data privacy and protection. The concern here is that artificial intelligence driven systems often require large datasets. These datasets often contain personal and sensitive data. With an AI system in possession of this data, “this raises potential conflicts with data protection regulations”. As in how the data is processed and accessed by the system.

Future Trends

Based on the research here, there are a few different ways that I think AI assisted red teaming could evolve over the next three to five years. The first thing is that with the AI that we have currently, human intervention will always be necessary. This is due to the fact that the advantages of AI being implemented is mostly for mundane tasks, which can allow the red teams to focus on the higher-level and more complex tasks at hand. Additionally, I can see the AI systems get efficient in terms of accuracy, as the more they evolve the better they will be at pattern recognition and identifying strategies. Also, I can see AI systems start to offer real time mitigation strategies in response to vulnerabilities or threats. I do not see AI leaving the cybersecurity realm now that it has entered. This is due to the fact that the advantages of implementing it far outweigh the disadvantages. Within the next 5 years I think that AI will be more implemented, with stricter regulations and rules in place, but also with human interaction

still. One of the biggest changes that I can see in the future is the legal and regulatory aspect. This is mostly due to the fact that there are a lot of grey areas with artificial intelligence, as it is still new and constantly evolving. Additionally, I do not think AI will go away, as it is heavily used by both offensive and defensive measures. One research question that I still have on the topic of artificial intelligence being used in pen testing is why there are not very many laws and regulations in place for it. Given that it is new and rapidly advancing, I thought there would be stricter laws and regulations. I think the main thing that practitioners and organizations should prepare for is automated defensive systems. This would allow them to focus on hiring people with the needed experience to expertly analyze those systems. This would also allow them to train their models and prioritize other vulnerabilities with the saved costs.

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